JONATHAN ALEXANDER YATES

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EDUCATION

Purdue University **Expected May 2029**
Bachelor of Science in Aerospace Engineering, Minor in Electrical & Computer Engineering West Lafayette, IN
GPA: 3.78 | **Honors:** Dean's List (2025, 2026), Semester Honors (2025, 2026)

TECHNICAL SKILLS

CAD & Analysis: Autodesk Fusion 360 (Certified CAD/CAM), Siemens NX (CAD & FEM), GD&T, Blueprint Reading
Controls & Software: Rockwell Studio 5000, RSLogix 500, FactoryTalk Optix Studio & View, GE iFIX, MATLAB, Python, C, C++, ROS
Fabrication & Hardware: CNC Mill CAM, Manual Lathe, GMAW/SMAW Welding (CTE Certified), Soldering, 3D Printing
Systems & Testing: Industrial Ethernet/IP Configuration, Avionics Hardware Packaging, Sensor Interfacing, Instrument Calibration

ENGINEERING & RESEARCH EXPERIENCE

NSG Pilkington **Summer 2026**
Electrical & Systems Engineering Intern Laurinburg, NC
- Programmed and troubleshot plant automation logic and safety interlocks utilizing Rockwell RSLogix 500 and Studio 5000 PLCs.
- Developed operator human-machine interfaces (HMI) and diagnostic alarm dashboards within Rockwell FactoryTalk Optix and GE iFIX.
- Configured industrial Ethernet/IP networks, subnet masks, and dedicated servers for machine-vision cameras and remote I/O drops.

Purdue MARS Lab - Guardian Rover **Fall 2026 - Present**
Undergraduate Researcher - Controls & Mechanical Design Purdue University
- Contribute to development of the Guardian rover control systems using C/C++ for microcontroller programming and Python for custom control scripts.
- Support development and testing of ROS-based rover software and integration between microcontrollers, control software, and hardware.
- Create and modify 3D CAD models and assemblies to support rover mechanical design and system integration.

Smithsonian MMT Observatory **Summer 2024**
Engineering Shadow Mount Hopkins, AZ
- Assisted lead staff engineers with mechanical maintenance, disassembly, and precision optical calibration on large-scale telescope assemblies.
- Observed diagnostic checkouts and servo motion control tuning for sub-arcsecond optomechanical positioning systems.

Thunderbolt Welding **2022 - 2024**
Fabricator & Welder Midland, NC
- Fabricated custom structural steel weldments, utility trailers, and fixtures adhering strictly to engineering blueprints and tolerances.
- Maintained shop compliance with OSHA-10 standards while safely operating horizontal band saws, ironworkers, and welding rigs.

ENGINEERING PROJECTS & LEADERSHIP

Purdue Space Program Active Controls - TOAD Lander Rocket **Aug 2025 - Present**
Structures Team Lead & Avionics Hardware Contributor Purdue University
- Direct mechanical structural design, load path analysis, and structural verification for an actively stabilized thrust-vectoring testbed rocket.
- Generate CNC toolpaths in Fusion 360 CAM and fabricate motor mounts and airframe brackets via CNC mills, manual lathes, and TIG/MIG welding.
- Coordinate with avionics teammates to integrate flight computers, thrust-vectoring gimbal servos, and custom harness packaging into chassis.

University of Arizona Solar Oven Engineering Challenge **2024**
Project Team Member & Challenge Winner (1st Place) Tucson, AZ
- Engineered, modeled, and tested a parabolic solar thermal oven, achieving 1st place in overall thermal concentration and structural design.

Purdue Scuba Club **Aug 2025 - Present**
Vice President Purdue University
- Direct organizational operations, manage club technical gear inspections, and coordinate logistics for dive excursions and safety training.

CERTIFICATIONS

- **Autodesk Certified User:** Fusion 360 CAD/CAM Design & Modeling
- **OSHA-10 General Industry Safety Certification:** Hazard recognition & safety protocols
- **CTE Welding & Fabrication Certification:** Blueprint reading, layout, & structural welding
- **SSI Advanced Open Water Diver:** Underwater navigation, gear rigging, & field safety